Symmetric PKM: repaired main and new native qualification

2026-10-30 · G1–G6 software prerequisite follow-up

All scientific acceptance remains open. G1–G3/G6 are partial; G4/G5 are unrun. Zero scientific panels, no admitted cost/GAE/scaling/global-novelty result, and comparison admission false. The full-panel card remains blocked/unfunded/unimplemented at its historical pin.

Reviewed repair and preservation

Copilot request 6531034718 had no response or visible source activity after more than one hour. The PR head/discussion were unchanged and the branch's in-progress Actions query returned no runs. Fallback explanation 6532293763 preceded the ordinary merge; no rebase or force push.

Merge b022566e0c6723fef4695a59e054ae4fbdcd1fd0 joins preserved local 270456262 and main cc5addeba4dd53edc51f2c0ad9331b3ba0307adc. Only daily-log and problem-index conflicts were resolved. All 1535/1681 parent paragraphs and 92/90 parent problem links are preserved, with 14 complete remote entry groups and the exact local prefix. The final index has 93 links. Main's hashed-ECDL/PQC rows and local PKM/JV/build rows survive. The prepared 18:54 request handoff was appended once after resolution.

No upstream change affected the PKM study or legacy controls. All frozen source/input/evidence/receipt files remain unchanged. The repaired chain-SAT blob is 4e2111e29f64c5fece73f2c52dd9aff9c6e7f69f; the JV source remains 568a67fb3a7e4067036c7b5985648c477a46290d. These are source identities, not a native compilation pass. The older a030 Cargo101/zero-test failure remains intact.

Exact formal equivalence

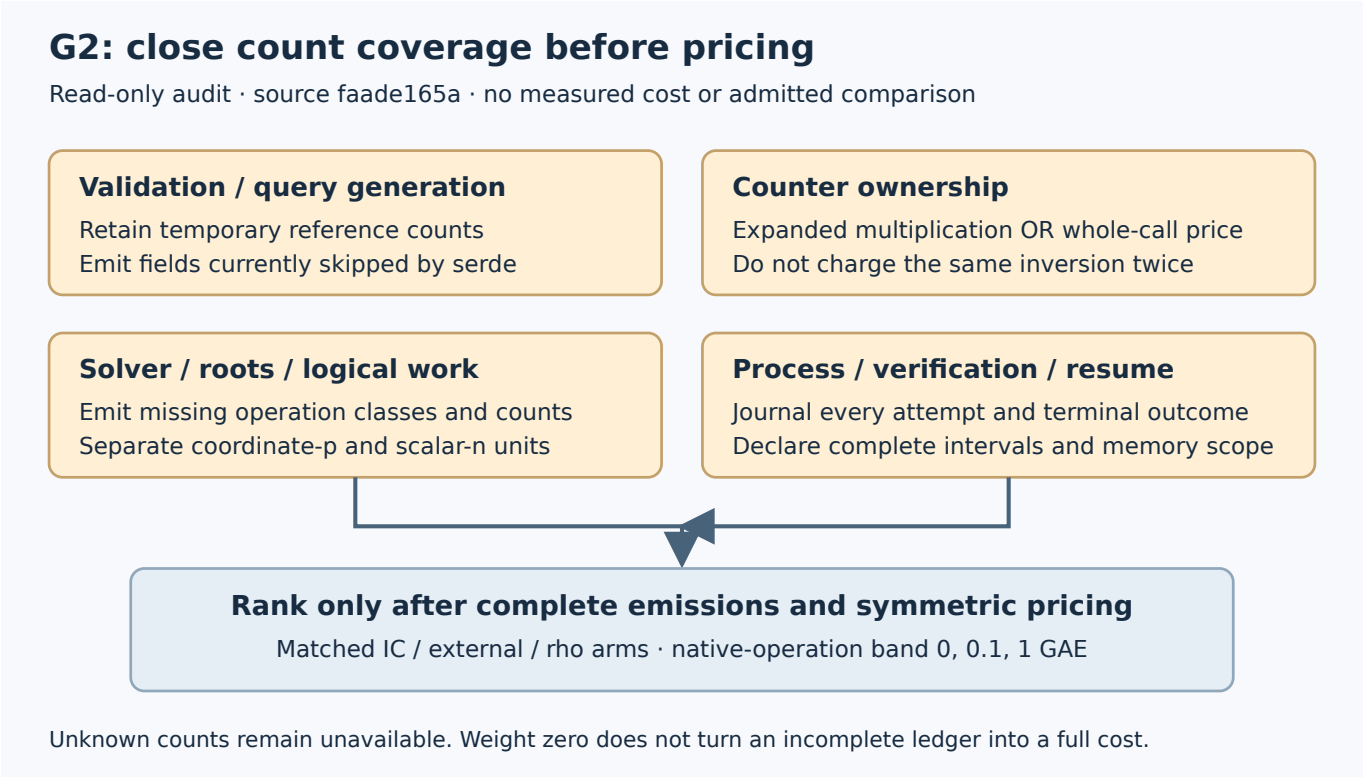
The project/toolchain/dependency and tools/lean gate objects exactly match the previously passing 18-declaration/zero-trusted receipt. Its official module replay is against imports, not –fresh. No redundant new formal run or Rust validation is claimed. The proof map's cyclic-group/cardinality, recovery-row and kernel hypotheses remain explicit; Rust/polynomial/complete-cost bridges remain open.

Fresh native run: pending, not passed

Source-pinned supervisor 49719 was live with Cargo 50095/rustc 50109 at 20:05:56UTC in example compilation. Exact recipe and 17 committed source/fixture bindings are in validation/repaired-main-cc5addeba/. Directory: /workspace/pkm-scalar-certificates-20261030/repaired-main-cc5addeba/. No interactive exec session exists. It checks source identity and tracked cleanliness before every phase. No bound-source edit or newer-main merge while active.

The sequence includes all 3 example suites, strict clippy/rustfmt, CLI build and actual checkpoint-refusal message/exit1, backlog build/validate, library build and a 2400-second serial library execution cap AFTER compilation. Actual phase outcomes, declared counts and binary names must be collected. Missing terminal/dead process is incomplete; wrapper 124 is not Cargo exit or a full-suite pass. The older 31539 and 46238 recipes are terminal and never restarted.

Old faade evidence remains source-scoped: 23 examples/checks pass there, while the 5646-case library run capped at 2208pass/0fail/47ignored/1active/3390not reached. Raw 214615 bytes SHA256 34fdd95ea133361298e8911f9a5899c431a5e9af97b73e7828e1b5289cdf87bd is NOT_UPLOADED. Earlier different-hash logs are distinct frozen records. None qualifies changed Rust. New raw logs over 64 KiB stay out of Git with exact final hash/bytes/path, bounded excerpt and NOT_UPLOADED if transport is unavailable; no compression/splitting or fake durable URI.



Scope after qualification

G2 still needs concrete count emissions and nonoverlapping ownership. Temporary validation counts are dropped, query field details are skipped by serialization, and inversion calls expand through counted multiplication helpers. Price the expanded leaves or calibrated whole call, avoiding duplicate charges. Solver/root/logical coverage, durable failed-call journaling, full process/replay/resume intervals and matched external/rho references remain incomplete. The audit and this diagram are not counter implementations or cost results.

Checkpoint/artifact/lab identities and a qualified source-bound native runtime/resume route remain unavailable. No scientific panel, funding, lease, IAM, quota or pool change occurred. The unchanged checkpoint/protocol/runtime gates remain in force. After current delivery, implement and test the missing G2 emissions before optimization; preserve the full G4/G5/G6 acceptance requirements.

Evidence and sources

See repair-20261030T2004Z.md, validation/main-cc5addeba-resolution.json and validation/repaired-main-cc5addeba/. Formal evidence is the exact-object-equivalent integration-0cc5/proof-receipt.json. Earlier failures/caps remain at their own pinned sources. Local commits are unpushed pending native qualification; auto-merge stays paused. Required exact-head gate,size,artifacts, applicable reviews/ resolved threads and a SHA guard remain mandatory before merge.

Primary PKM 2016 Algorithm 1/§3.2 includes permutation symmetry, relation dependence and linear algebra: people.maths.ox.ac.uk/petit/files/16PKC_primeECDLP.pdf. The retained novelty audit, actual native/Lean source and G2_ACCOUNTING_AUDIT.md supply scope. No new mathematical identity, algorithm or performance measurement is introduced. The diagram is explanatory because there are no comparable admitted measurements; canonical IC/curve/performance graphs stay unchanged. Old reports/PDFs are preserved. Software integration does not complete G1–G6.